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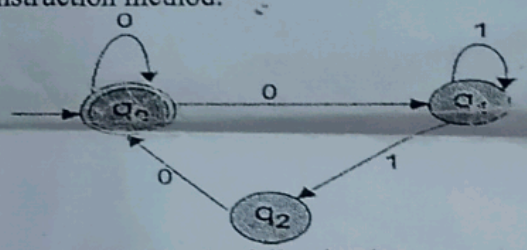
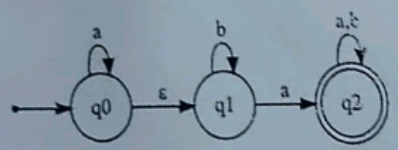
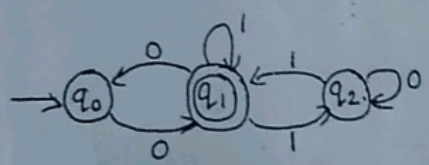
Vivekananda College of Engineering & Technology, Puttur
[A Unit of Vivekananda Vidyavardhaka Sangha Puttur ®]
Affiliated to VTU, Belagavi & Approved by AICTE New Delhi

2025-26 (Odd)	CRM08	Rev 1.17 (2025 rev)	CD	23/09/25

CONTINUOUS INTERNAL EVALUATION- 1

Dept: AI/CD/CS	Sem : 5th	**Sub: Theory of Computation**	**S Code:BCS503**
Date:29/09/25	Time: 9:30-11:00 am	Max Marks: 50	Elective: N

Note: Answer any 2 full questions, choosing one full question from each part.

Q N		Questions	Marks	RBT	COs
		PART A			
1	a	Define DFA and Construct DFA for the following and verify your answer by taking one example string. i) $L = \{w \in \{a, b\}^* : w$ is of odd length and begins with **'a'**$\}$ ii) $L = \{w \in \{a\} : \lvert w\rvert \bmod 3 = \lvert w\rvert \bmod 2\}$ iii) $L = \{w \in \{0,1\}^* :$ strings having even 0's, odd 1's $\}$	10	L3	CO1
	b	i) Convert the following NFA to DFA using subset construction method.  ii) Design a NFA for the following language: $L = \{\{0101^n \mid n>=1\}$ or $\{011^n \mid n>=0\}\}$	10	L3	CO1
	c	Define CFG with example. Obtain CFG for $L = \{L=a^n b^{n-2} \mid n \geq 2\}$. Take one example to justify your grammar.	5	L2	CO3
		OR			
2	a	Define ECLOSE. Convert the following ε- NFA to DFA by finding ECLOSE of all states.  	10	L3	CO1
	b	i) Convert the following NFA to DFA using lazy evaluation method.  ii) Define Powers of alphabet (Σ^*) with example.	10	L3	CO1
	c	Obtain CFG to generate a language of 0's and 1's having substring 010. Take one example to justify your grammar. What is the language generated by the grammar shown below. $S \rightarrow aA \mid \varepsilon$ $A \rightarrow bS$	5	L2	CO3

Prepared by: HOD

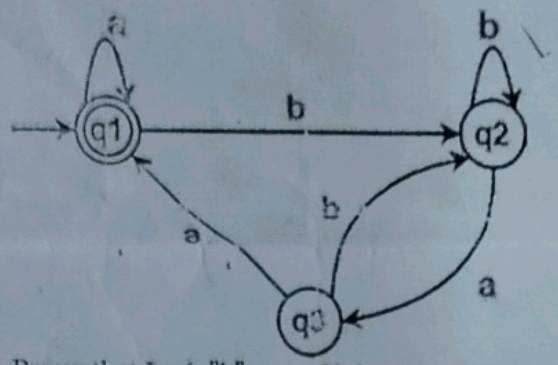
CONTINUOUS INTERNAL EVALUATION- 1

PART B

3	a	Minimize the following DFA using table filling algorithm.	10	L3	CO2

δ	a	b
→A	B	F
B	G	C
*C	A	C
D	C	G
E	H	F
F	C	G
G	G	E
H	G	C

	b	i) Convert the following FA to RE using State elimination method.	10	L3	CO2



ii) Prove that $L=\{a^n b^n : n>=0\}$ is not regular using pumping lemma.

	c	Obtain LMD, RMD and Parse Tree for the string " (a,a)" using CFG S→ (L) | a L→ L , S | S	5	L2	CO3

OR

4	a	Define Regular Expression and Write RE for the following: i) Strings of a's and b's having both 'aa' and 'bb' as substrings. ii) $L= \{a^n b^m \mid n>=4, m<=3 \}$ iii) $L=\{w \in \{0\text{-}9\}^* : w$ *is* an odd natural number$\}$. iv) Strings of a's and b's not ending with 'bb'	10	L3	CO2
	b	Convert the following RE to FA. i) ab (a+b)* ii) a* + b* + c*	10	L3	CO2
	c	Show that the following CFG is ambiguous for the string "(( ))( )" S → (S) | SS | ε	5	L2	CO3

Prepared by: Roopa G.K. 23/09/25 24/09/25 HOD 23/09/25